

The Missing Order Mystery (Case Study)

1. System Design

A) Current Flow (As-Is)

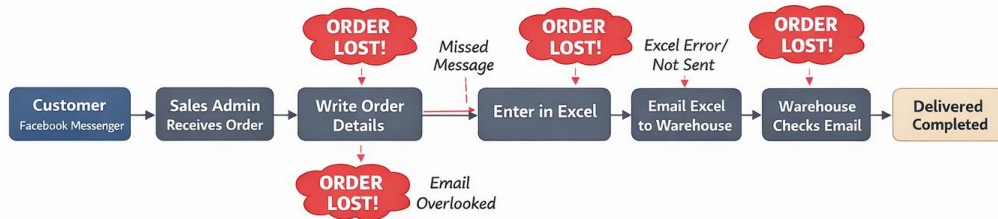
Current Process Steps:

1. Customer places order via Facebook Messenger.
2. Sales Admin reads and manually records order details.
3. Sales Admin enters order into Excel.
4. Excel file is emailed to Warehouse.
5. Warehouse checks email and processes order.

Common Failure Points:

- Missed messages
- Data entry errors in Excel
- Email not sent or overlooked
- No tracking or status visibility

Current Flow Diagram



Failure risk exists between each manual step.

B) Proposed System Flow (Web App + Database)

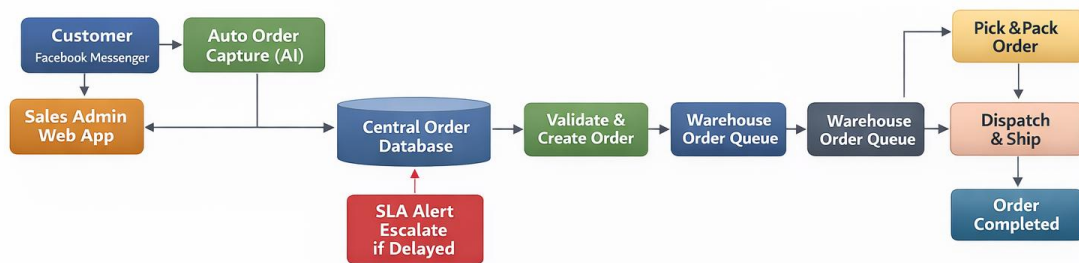
Proposed Improvements:

- Centralized Web Application
- Real-time Database (Single Source of Truth)
- Role-based dashboards (Sales / Warehouse)
- Automatic Notifications
- SLA Monitoring & Escalation
- AI-assisted Order Extraction

Proposed Flow Steps:

1. Customer message received via Messenger API/Webhook.
2. System captures and stores raw message in database.
3. AI extracts structured order details.
4. Sales Admin reviews and confirms order.
5. Order saved with status = NEW.
6. Warehouse receives instant notification.
7. Warehouse updates status (ACKNOWLEDGED → PACKED → SHIPPED → COMPLETED).

Proposed Flow Diagram



2. AI & Logic Check

System Structure

Frontend:

- Sales Dashboard (Create/Review Orders)
- Warehouse Dashboard (Process Orders)

Backend:

- Authentication Service
- Order Management API
- AI Extraction Service
- Notification Service
- SLA Monitoring Service

Database Tables (Example)

- users (id, name, role)
- orders (id, customer_name, phone, address, items, status)
- order_status_history (order_id, old_status, new_status)
- messages (raw_text, received_at)
- notifications (order_id, channel, status)

Core Logic Rules

- All orders must exist in the database before processing.
- Status transitions must follow workflow rules.
- SLA timer triggers escalation if warehouse does not acknowledge in time.
- All actions are logged for audit trail.

AI Integration

Example AI Prompt:

Extract structured order data from the message and return JSON with:
customer_name, phone, address, items, payment_method, delivery_type, missing_fields.

AI Workflow Explanation:

1. Raw message stored.
2. AI extracts structured draft data.
3. Sales Admin reviews AI output.
4. Confirmed order becomes official in database.

AI assists but does not replace human validation.

3. IT Attitude (Scenario Handling)

If system failure occurs:

- Step 1: Identify scope of issue (Database, API, Notification, AI).
Step 2: Activate fallback process (temporary manual entry form).
Step 3: Restore services or roll back recent changes.
Step 4: Reconcile missed orders using message logs.
Step 5: Conduct Root Cause Analysis and implement monitoring.

Professional Mindset:

- Focus on business continuity.
- Communicate clearly with stakeholders.
- Document incident and prevent recurrence.